

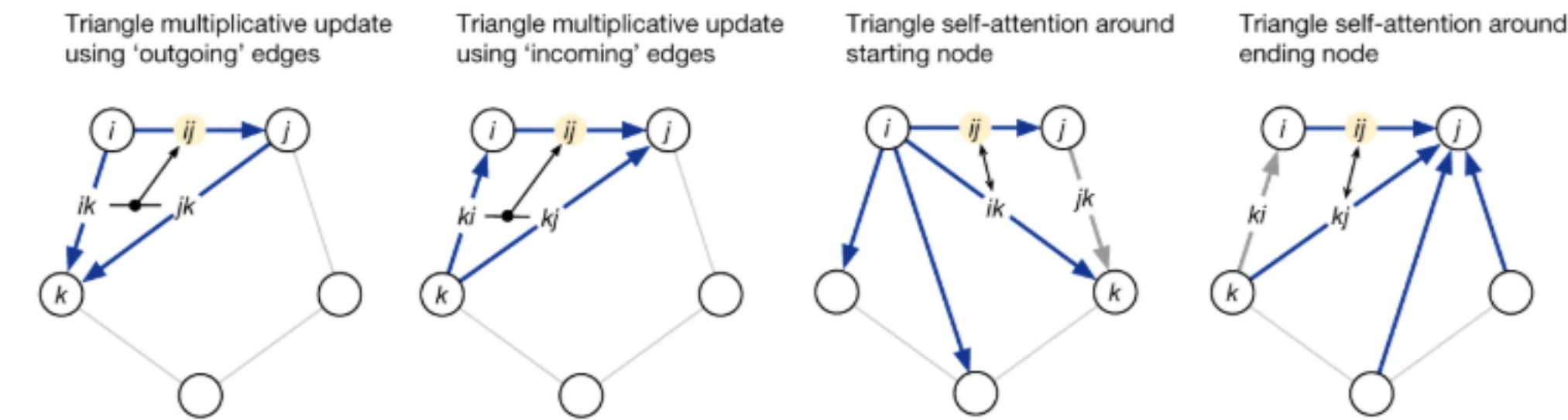
Simplicial complex CNNs

Node-edge-triangle interactions

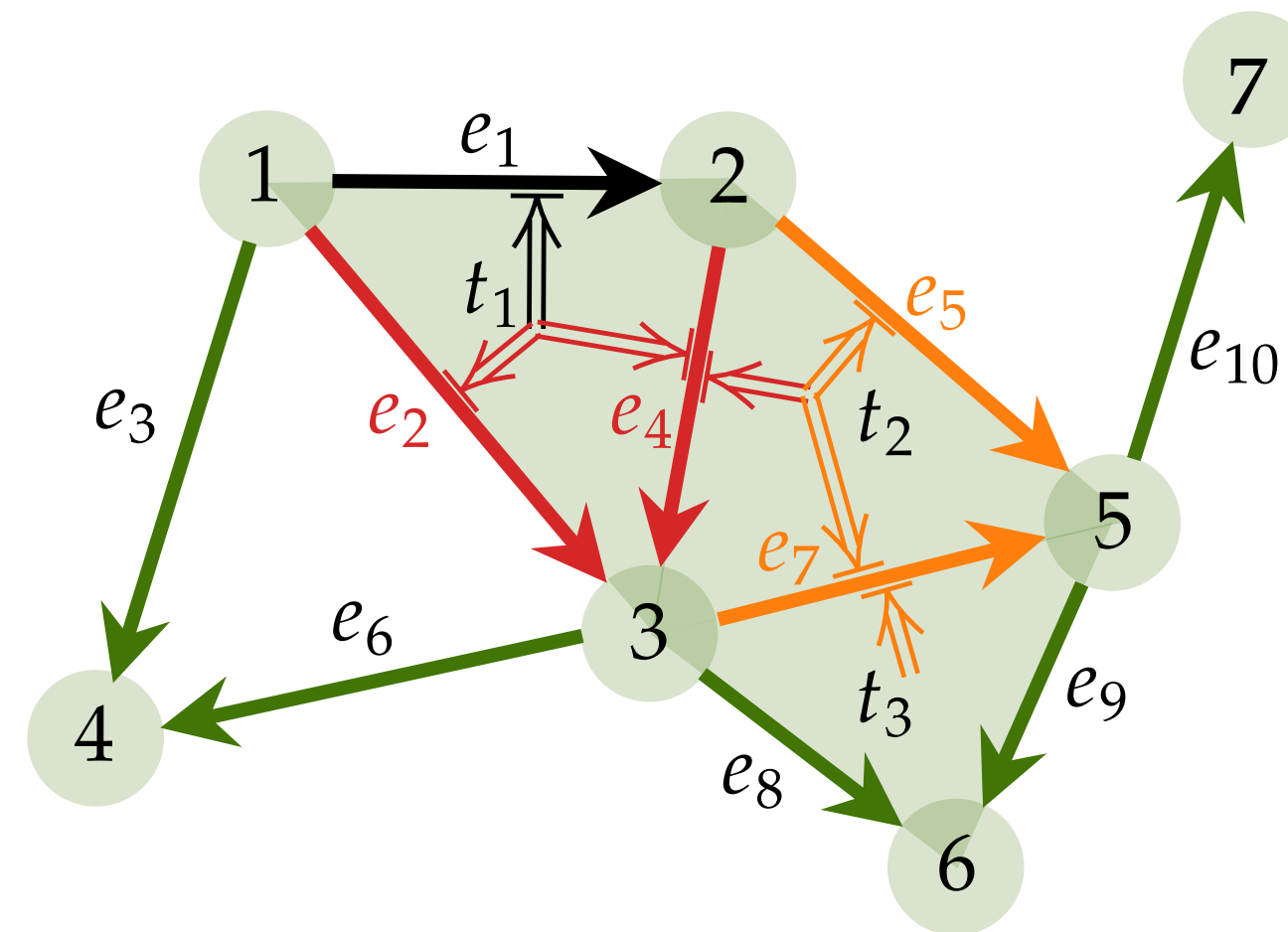
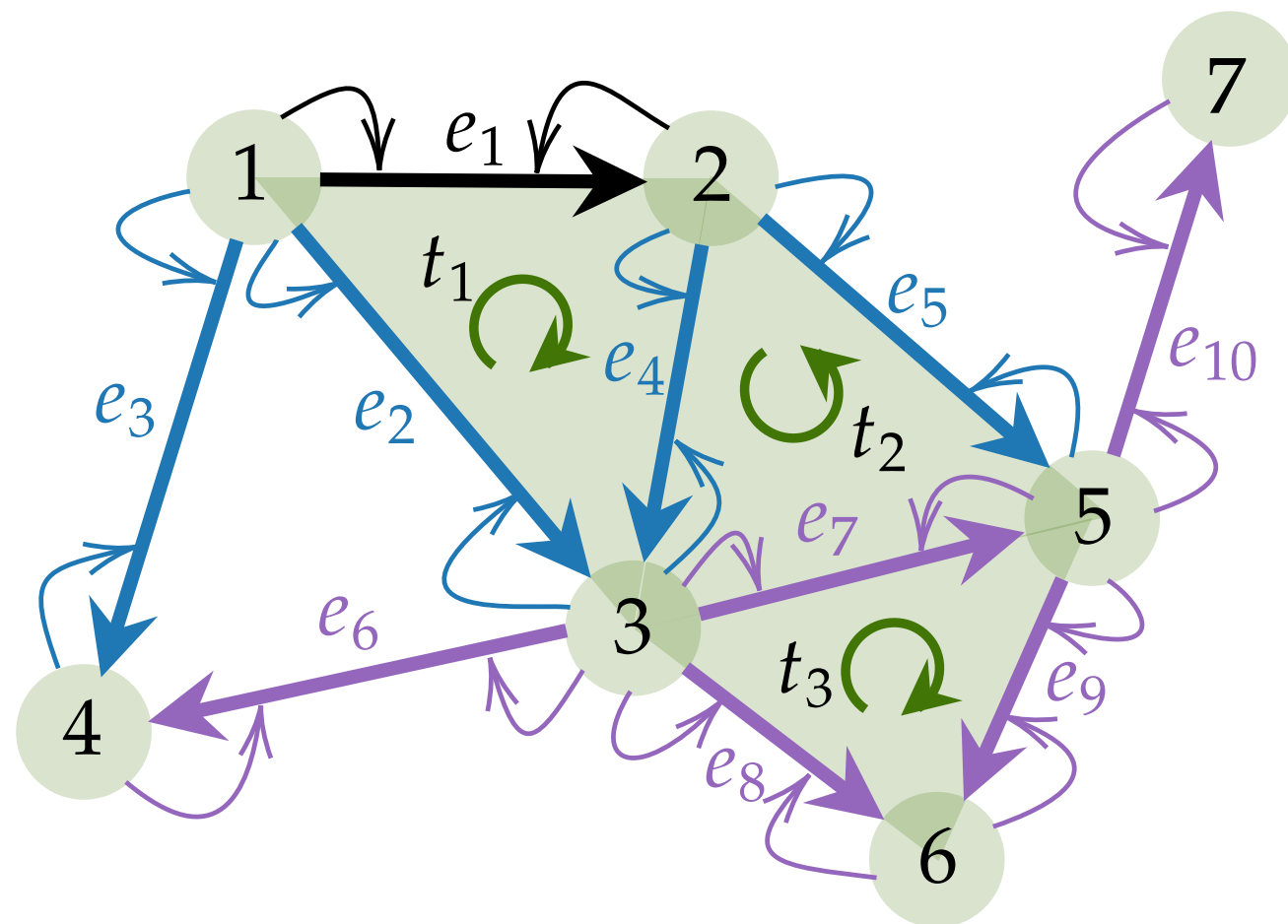
Node $\leftarrow$ node, edge
 Edge $\leftarrow$ node, edge, triangle
 Triangle $\leftarrow$ edge, triangle

Chebnet

$$\begin{aligned} \mathbf{x}_0^l &= \sigma(\mathbf{H}_0^l \mathbf{x}_0^{l-1} + \mathbf{H}_{0,u}^l \mathbf{B}_1 \mathbf{x}_1^{l-1}) \\ \mathbf{x}_1^l &= \sigma(\mathbf{H}_{1,d}^l \mathbf{B}_1^\top \mathbf{x}_0^{l-1} + \mathbf{H}_1^l \mathbf{x}_1^{l-1} + \mathbf{H}_{1,u}^l \mathbf{B}_2 \mathbf{x}_2^{l-1}) \\ \mathbf{x}_2^l &= \sigma(\mathbf{H}_{2,d}^l \mathbf{B}_2^\top \mathbf{x}_1^{l-1} + \mathbf{H}_2^l \mathbf{x}_2^{l-1}) \end{aligned}$$



AlphaFold2 paper Fig 3.
 Core element in AlphaFold3/4



Properties: locality, symmetry
 Dirichlet energy perspective
 Hodge-invariant
 Stability to weights perturbations
 Extensions:

- message-passing
- Attention
- E(3)-equivariance

Convolution based (Ebli et al. 2020; Roddenberry et al. 2021; Yang et al. 2022, 2023)
 Message passing (Bodnar et al. 2021)

Simplex prediction

Generalization of link prediction

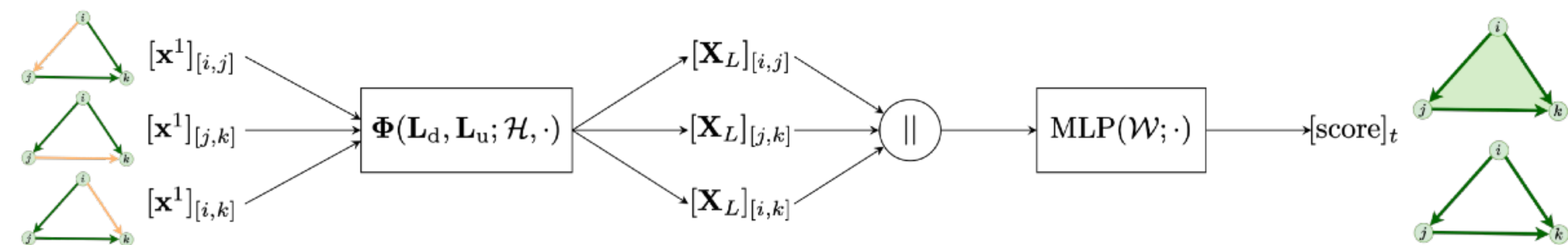


Table 2: Simplex prediction (AUC, $\uparrow$) .

Methods	2-simplex	3-simplex
Mean (Benson et al., 2018)	62.8±2.7	63.6±1.6
MLP	68.5±1.6	69.0±2.2
GNN (Defferrard et al., 2016)	93.9±1.0	96.6±0.5
SNN (Ebli et al., 2020)	92.0±1.8	95.1±1.2
PSNN (Roddenberry et al., 2021)	95.6±1.3	98.1±0.5
SCNN (Yang et al., 2022a)	96.5±1.5	98.3±0.4
Bunch (Bunch et al., 2020)	98.3±0.5	98.5±0.5
MPSN (Bodnar et al., 2021b)	98.1±0.5	99.2±0.3
SCCNN	98.7±0.5	99.4±0.3

Chebnet

Model	COVID	Ribonanza	Tc-Ribo
<i>1D model</i>			
Transformer1D	0.361±0.017	0.705±0.015	0.705±0.079
RNA-FM	0.591±0.081	0.990±0.144	0.693±0.001
SpliceBERT	0.588±0.077	1.022±0.144	0.708±0.003
<i>2D model</i>			
Transformer1D2D	0.305±0.012	0.514±0.004	0.633±0.001
GCN	0.359±0.009	0.595±0.006	0.701±0.004
GAT	0.315±0.006	0.534±0.006	0.685±0.024
ChebNet	0.279±0.007	0.468±0.002	0.621±0.022
Graph Transformer	0.318±0.008	0.515±0.001	0.710±0.041
GraphGPS	0.332±0.013	0.523±0.003	0.715±0.012
<i>3D model (without pooling)</i>			
EGNN	0.480±0.025	0.808±0.023	0.725±0.002
SchNet	0.499±0.003	0.843±0.004	0.696±0.008
FAENet	0.486±0.010	0.834±0.003	0.703±0.011
DimeNet	0.497±0.012	0.855±0.006	0.712±0.004
GVP	0.467±0.010	0.797±0.012	0.744±0.004
FastEGNN	0.477±0.005	0.816±0.014	0.753±0.001
<i>3D model (with nucleotide pooling)</i>			
EGNN (pooling)	0.364±0.003	0.619±0.007	0.663±0.010
SchNet (pooling)	0.390±0.006	0.685±0.006	0.655±0.038
FastEGNN (pooling)	0.444±0.003	0.753±0.015	0.710±0.011

- Others:
- Molecule property prediction
 - E3-equi. extension for n-body systems